EXPERIMENT-1

ADDITION:

AIM:

To prove the program for addition using R-tool.

PROGRAM:

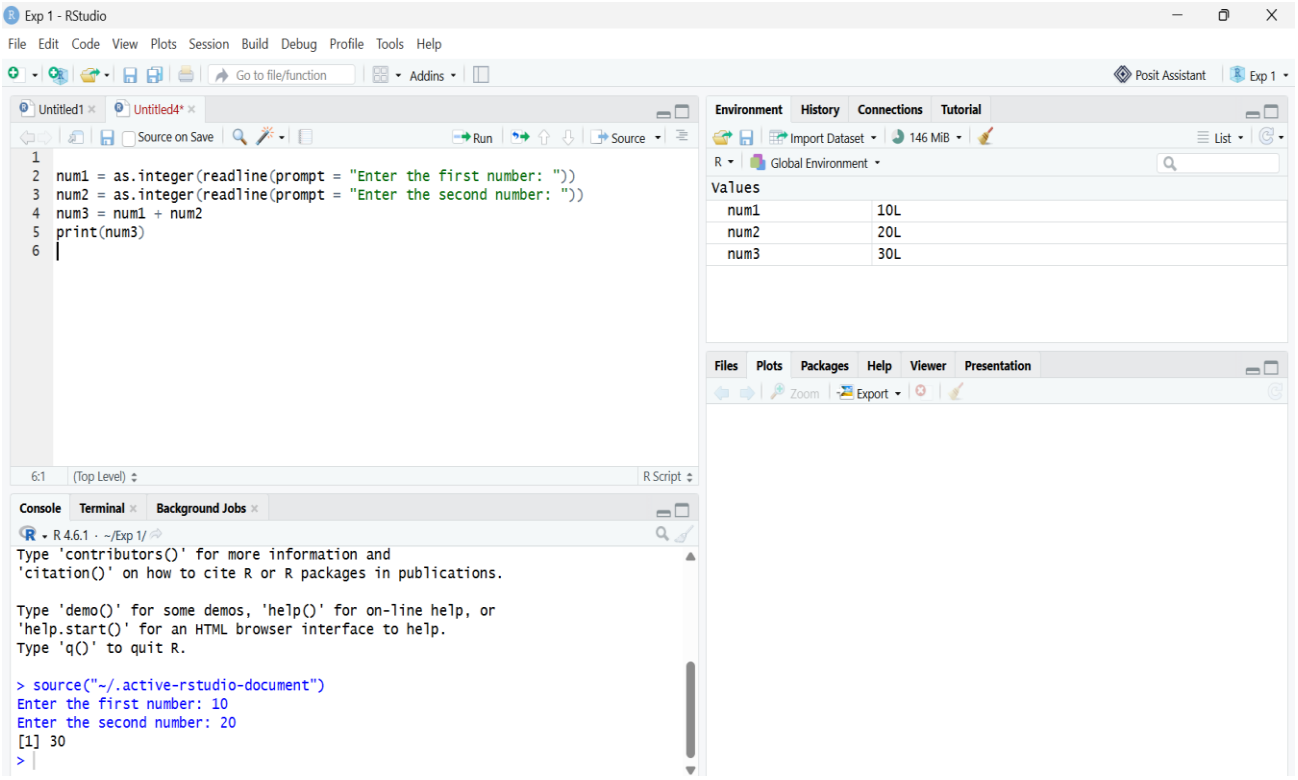
num1=as.integer(readline(prompt = “enter the first number:”))

num2=as.integer(readline(prompt = “enter the second number:”))

num3=num1+num2

print(num3)

OUTPUT:



RESULT:

Thus the basic program addition is executed successfully.

EXPERIMENT-2

SUBTRACTION:

AIM:

To prove the program for subtraction using R-tool.

PROGRAM:

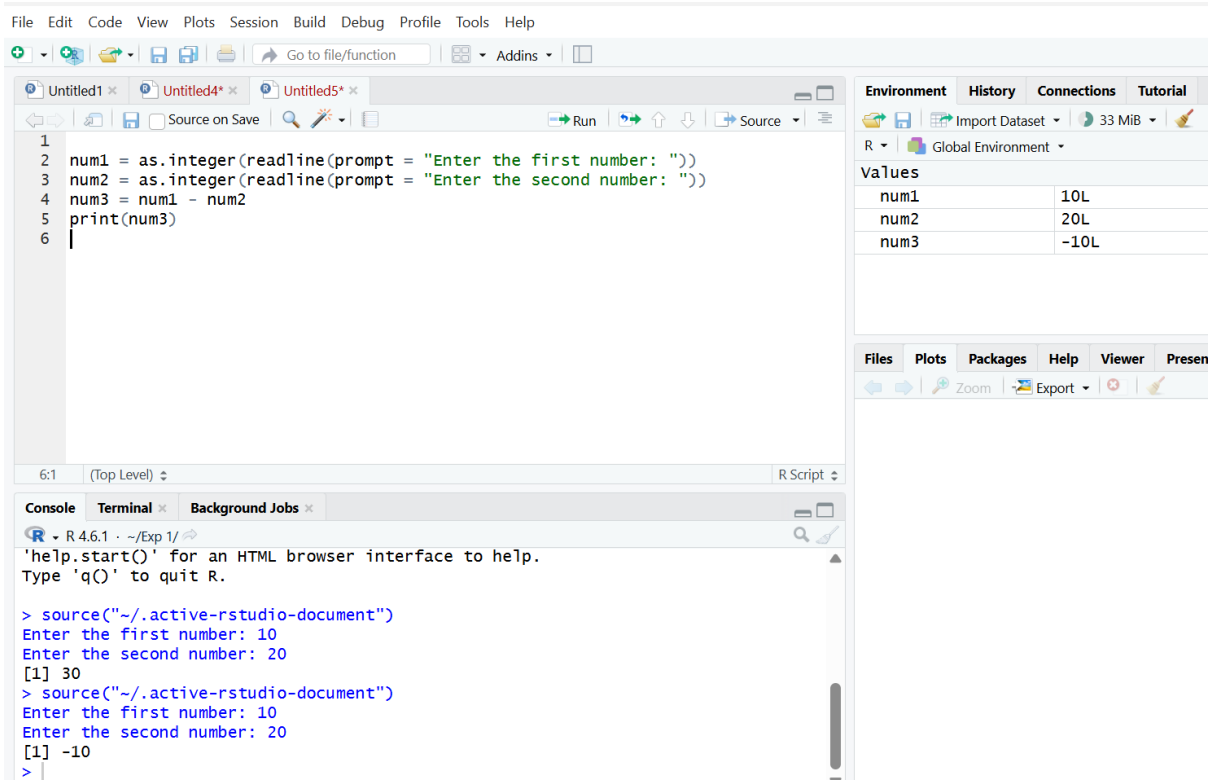
num1=as.integer(readline(prompt = “enter the first number:”))

num2=as.integer(readline(prompt = “enter the second number:”))

num3=num1-num2

print(num3)

OUTPUT:



RESULT:

Thus the basic program subtraction is executed successfully.

EXPERIMENT-3

MULTIPLICATION:

AIM:

To prove the program for multiplication using R-tool.

PROGRAM:

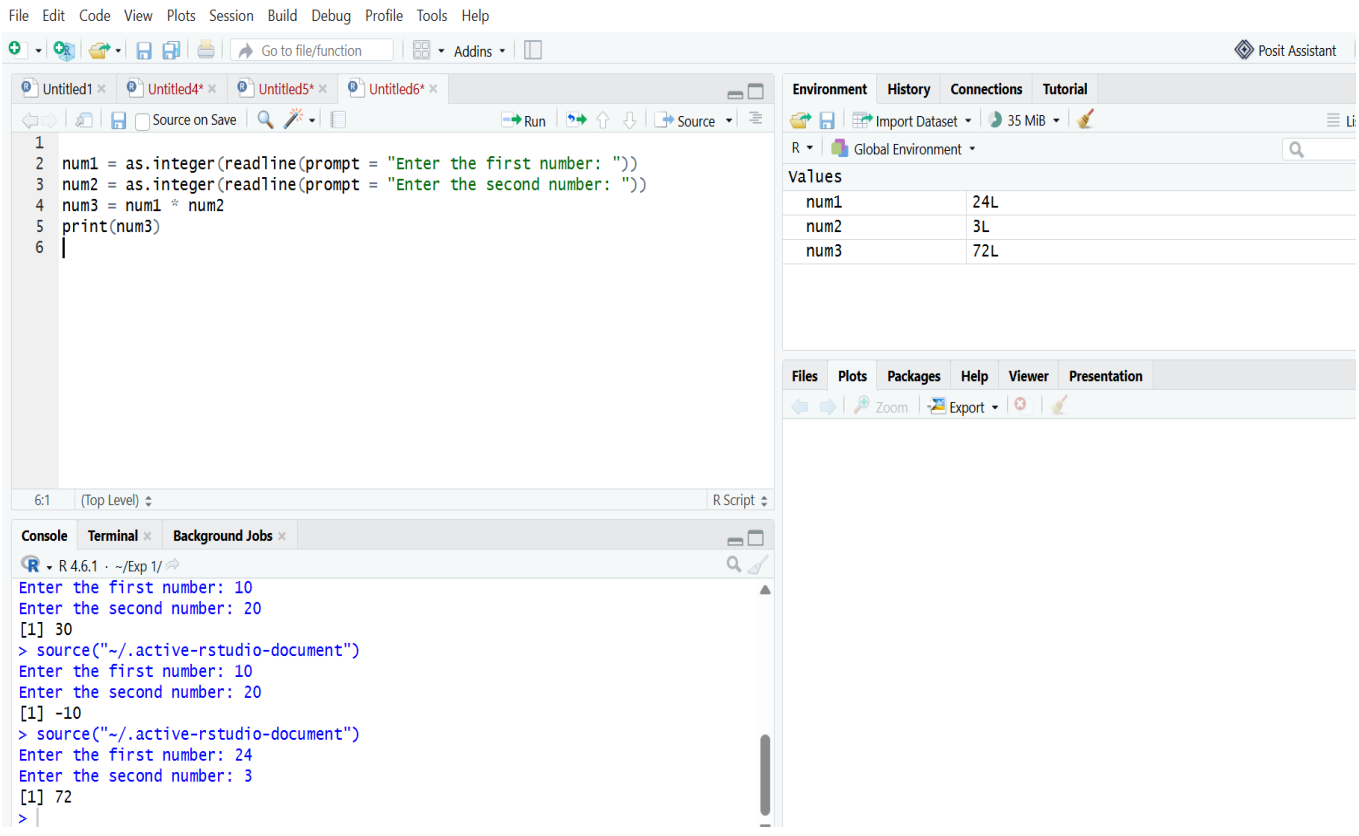
num1=as.integer(readline(prompt = “enter the first number:”))

num2=as.integer(readline(prompt = “enter the second number:”))

num3=num1*num2

print(num3)

OUTPUT:



RESULT:

Thus, the basic program multiplication is executed successfully.

EXPERIMENT-4

DIVISION:

AIM:

To prove the program for division using R-tool.

PROGRAM:

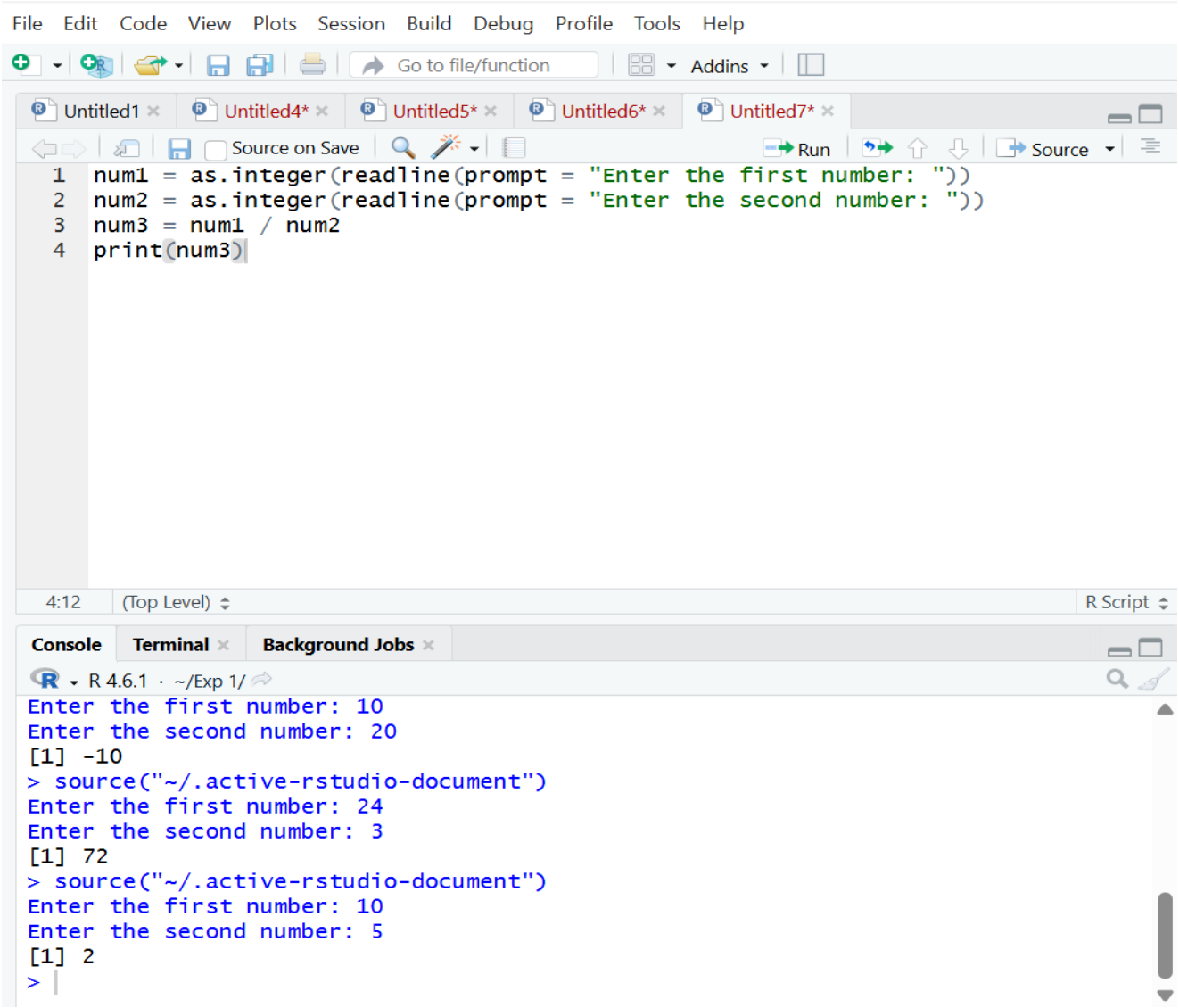
num1=as.integer(readline(prompt = “enter the first number:”))

num2=as.integer(readline(prompt = “enter the second number:”))

num3=num1/num2

print(num3)

OUTPUT:



RESULT: Thus, the basic program division is executed successfully.

EXPERIMENT-5

AIM:

To write the program for odd or even using R-tool.

PROGRAM:

num=as.integer(readline(prompt="enter a number:"))

if((num%%2)==0)

{

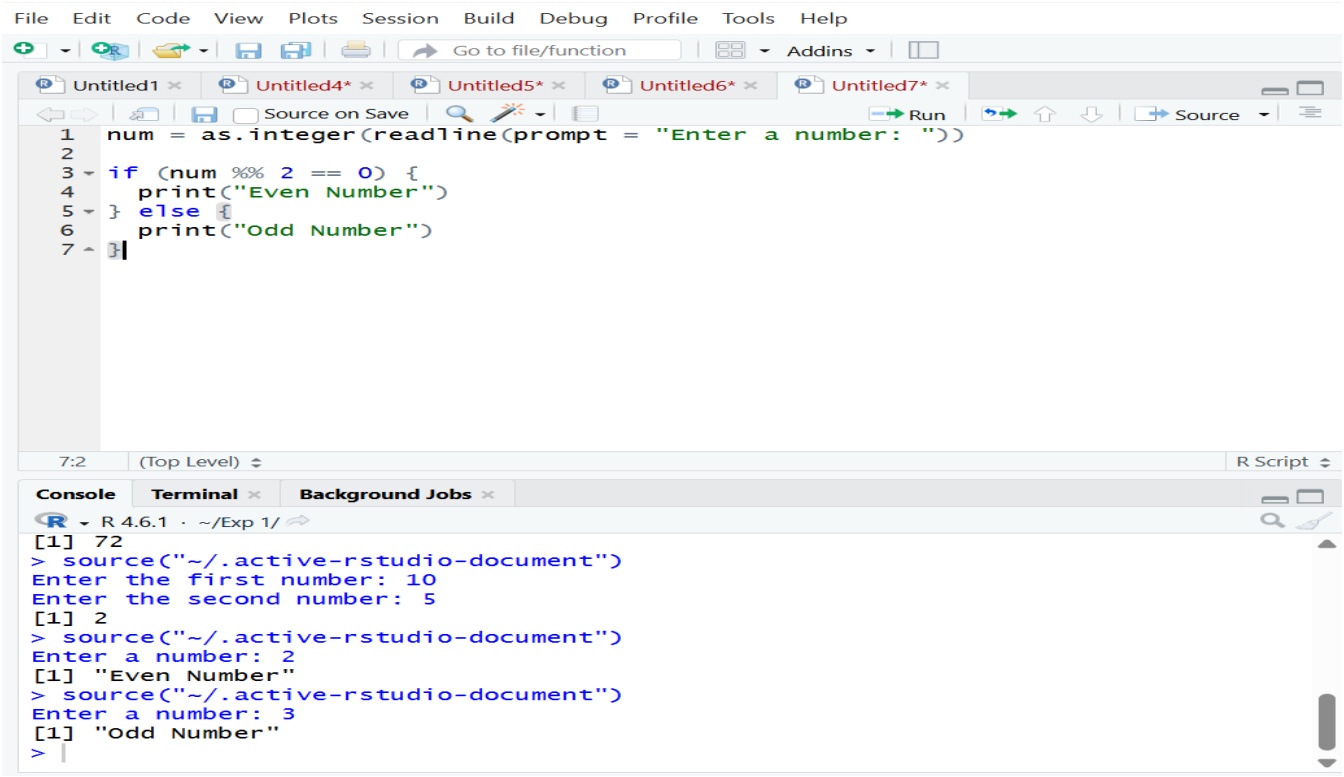
print("number is a even")

}else{

print("number is odd")

}

OUTPUT:



RESULT: Thus, the basic program Odd or Even is executed successfully.

EXPERIMENT-6

AIM:

To write the program for mean,median,mode.

PROGRAM:

MEAN

```
names&lt;-c(&quot;siri&quot;,&quot;mahi&quot;,&quot;chiru&quot;)
age&lt;-c(23,24,25)
marks&lt;-c(88,78,25)
df&lt;-data.frame(names,age,marks)
mean(df $age)
write.csv(df,&quot;datafr.csv&quot;)
```

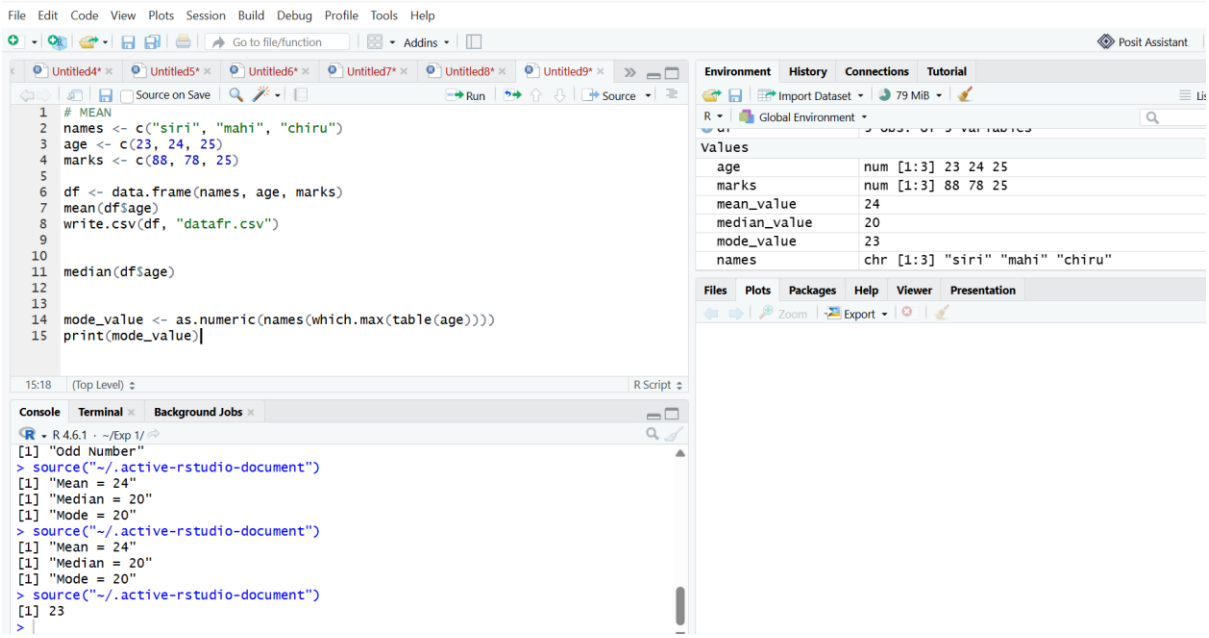
MEDIAN

```
names&lt;-c(“siri”,&”mahi”;”chiru”)
age&lt;-c(23,24,25)
marks&lt;-c(88,78,25)
df&lt;-data.frame(names,age,marks)
median(df $age)
write.csv(df,&quot;datafr.csv&quot;)
```

MODE

```
names&lt;-c(&quot;siri&quot;,&quot;mahi&quot;,&quot;chiru&quot;)
age&lt;-c(23,24,25)
marks&lt;-c(88,78,25)
df&lt;-data.frame(names,age,marks)
mode(df $age)
write.csv(df,&quot;datafr.csv&quot;)
```

OUTPUT:



RESULT: Thus, the central tendancy and measure of dispersion is executed successfully.

EXPERIMENT-7

AIM:

To write the program for summary using R-tool.

PROGRAM:

names<-c("siri","mahi","chiru")

age<-c(23,24,25)

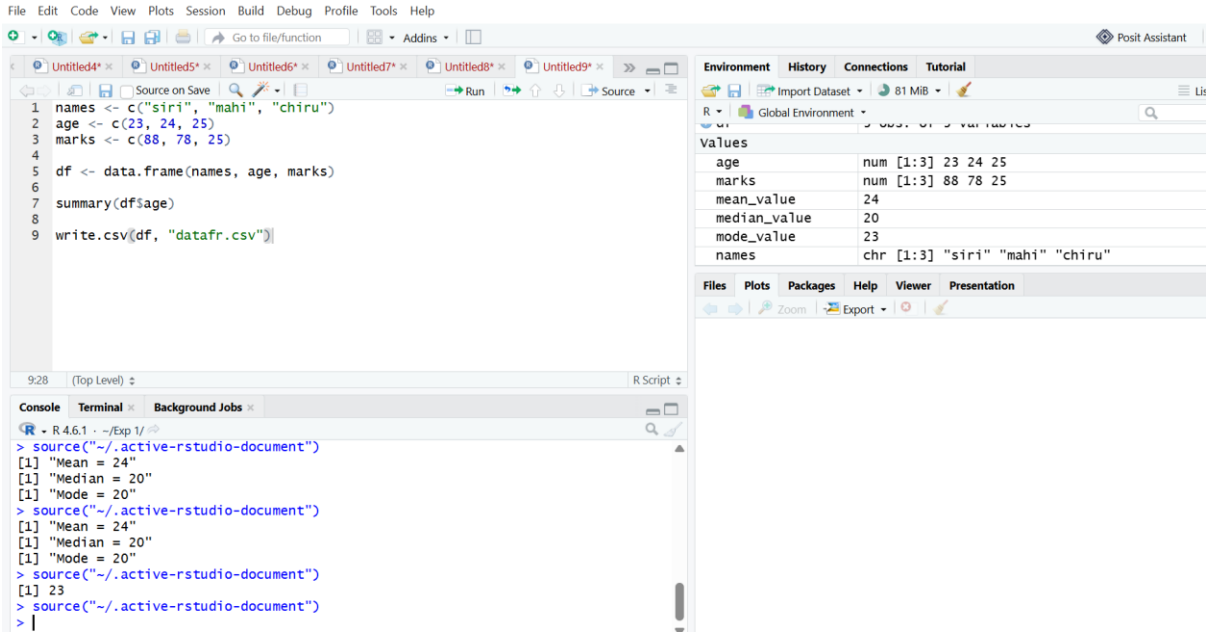
marks<-c(88,78,25)

df<-data.frame(names,age,marks)

summary(df $age)

write.csv(df,"datafr.csv")

OUTPUT:



RESULT: Thus, the central tendancy and measure of dispersion is executed successfully.

EXPERIMENT 8 : GREATER AMONG THREE NUMBERS

AIM:

To write the program for the greatest among three numbers.

PROGRAM:

x <- as.integer(readline(prompt = "Enter first number :"))

y <- as.integer(readline(prompt = "Enter second number :"))

z <- as.integer(readline(prompt = "Enter third number :"))

if (x > y && x > z) {

print(paste("Greatest is :", x))

} else if (y > z) {

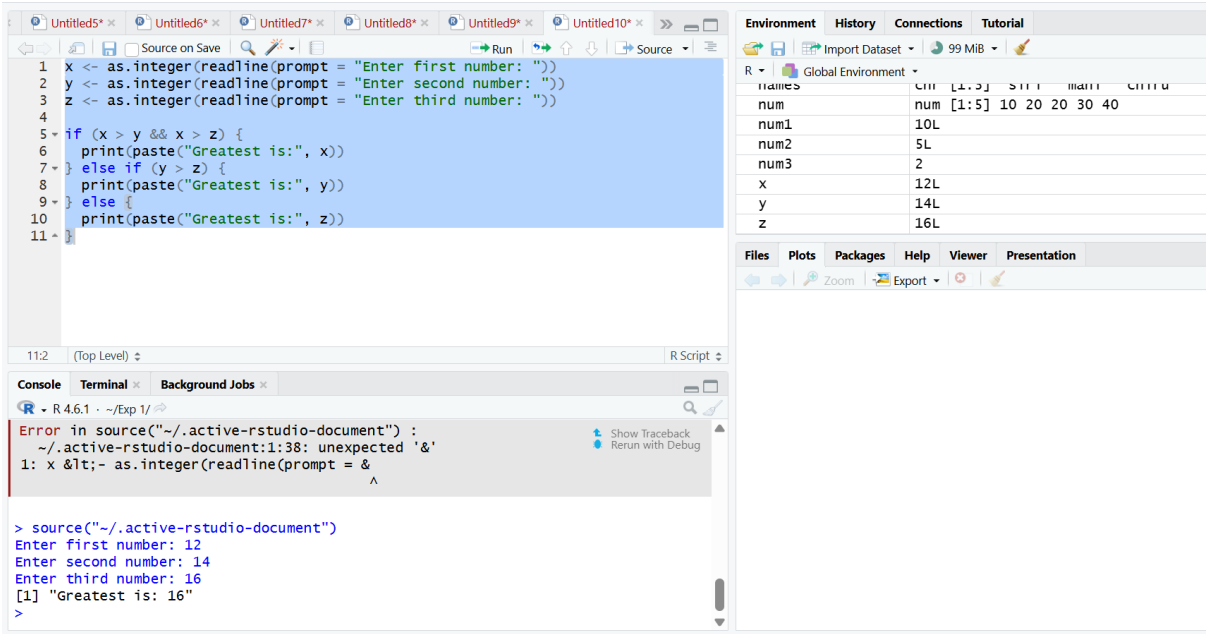
print(paste("Greatest is :", y))

} else{

print(paste("Greatest is :", z))

}

OUTPUT:



RESULT:

Thus, the greatest among the three numbers was executed successfully.

EXPERIMENT-9 IQR

AIM:

To write the program for central tendency and data dispersion measures using R tool.

PROGRAM:

names<-c("siri","mahi","chiru")

age<-c(23,24,25)

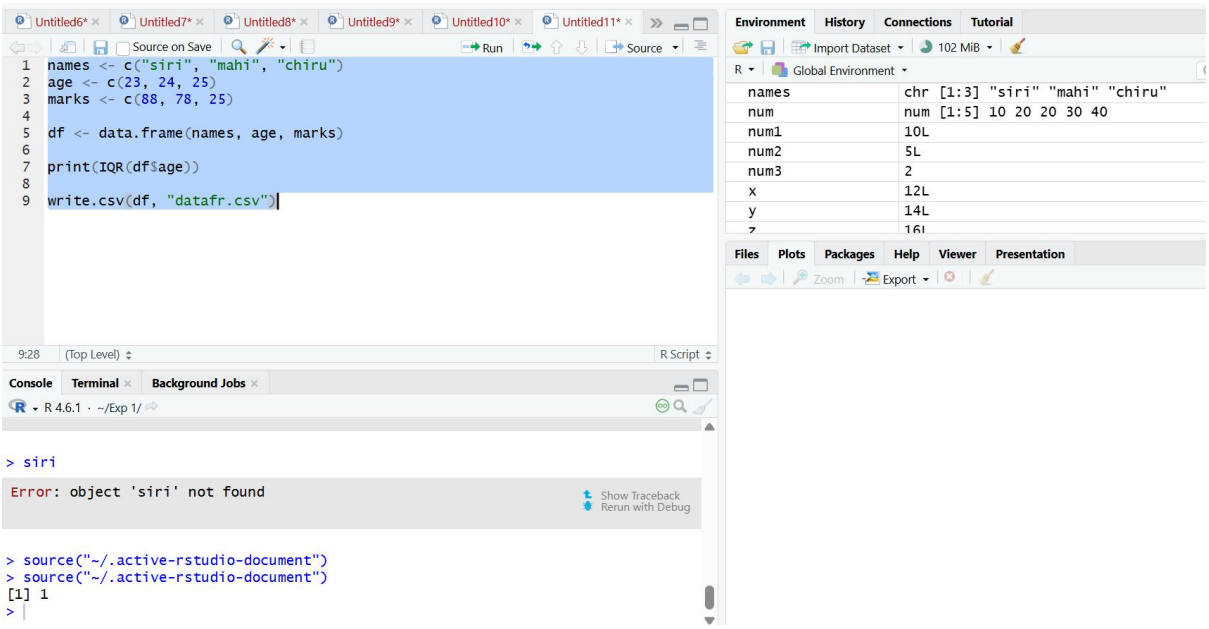
marks<-c(88,78,25)

df<-data.frame(names,age,marks)

IQR(df $age)

write.csv(df,"datafr.csv")

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully.

EXPERIMENT-10 QUANTILE

AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

names<-c("siri","mahi","chiru")

age<-c(23,24,25)

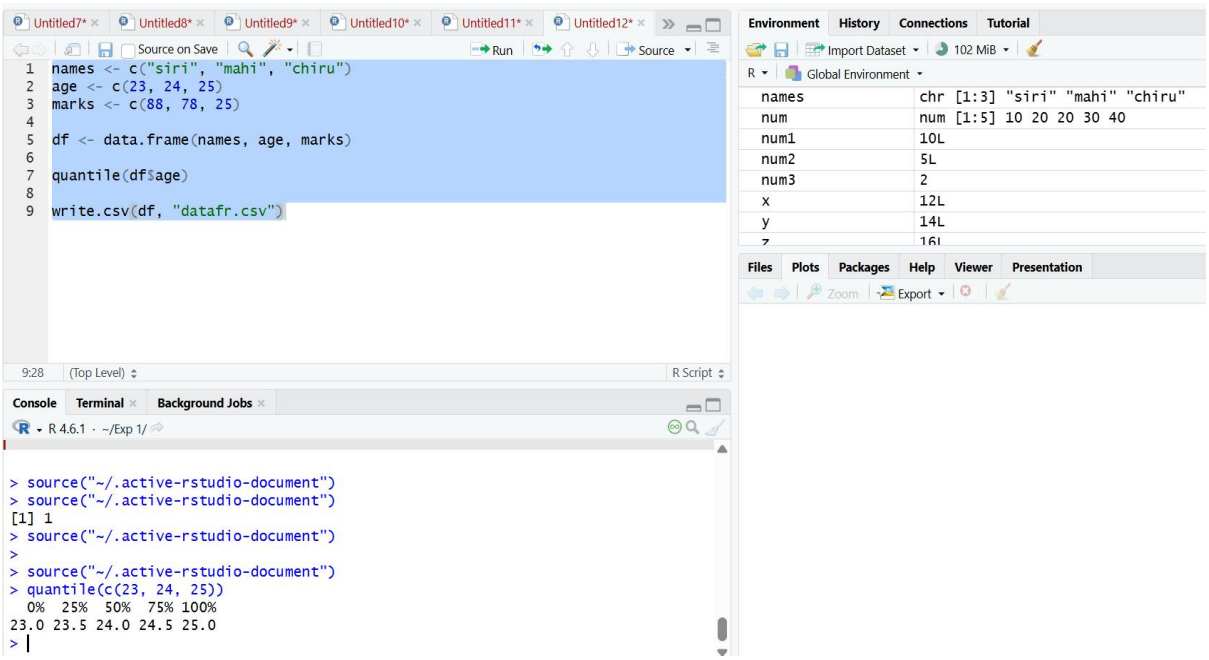
marks<-c(88,78,25)

df<-data.frame(names,age,marks)

quantile(df $age)

write.csv(df,"datafr.csv")

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully

EXPERIMENT-11 MID RANGE:

AIM:

To write the program for central tendency and data dispersion measures.

PROGRAM:

names<-c("siri","mahi","chiru")

age<-c(23,24,25)

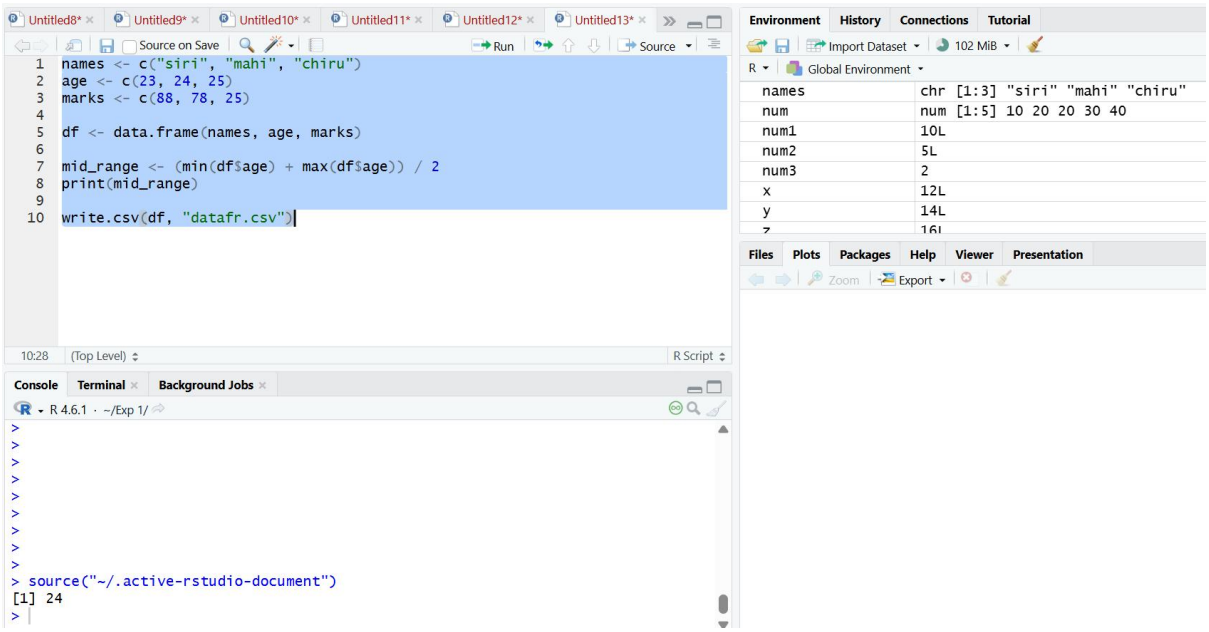
marks<-c(88,78,25)

df<-data.frame(names,age,marks)

mid range(df $age)

write.csv(df,"datafr.csv")

OUTPUT:



RESULT:

Thus, the program for central tendency and data dispersion measures was executed successfully

EXPERIMENT-12 Z-SCOORE NORMALIZATION:

AIM:

To write the program for Z-scoore normalization using R-tool.

PROGRAM:

diabetest1<-read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")

A<-c(diabetest1$Age)

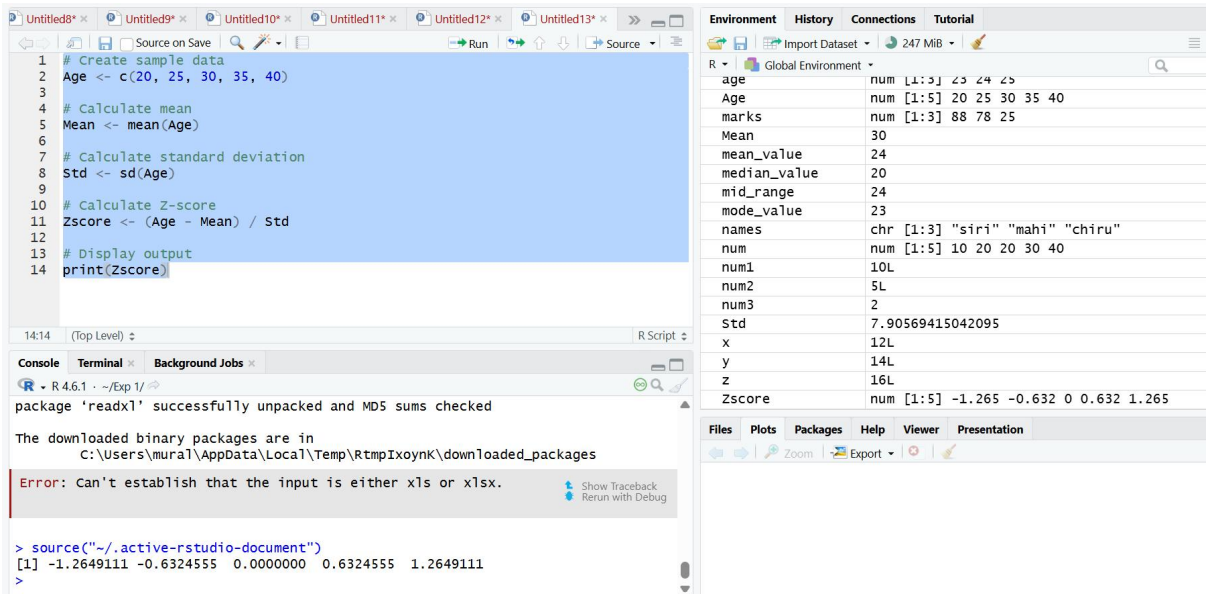
Mean<-mean(A)

Std<-sd(A)

Zscore<-(A-Mean)/Std

Zscore

OUTPUT:



RESULT:

Thus, the Z-score normalization using R tool was executed successfully.

EXPERIMENT-13 MIN,MAX,MEAN,MINMAX

AIM:

To write the program for the minimum,maximum,mean and minmax using r-TOOL

PROGRAM:

MEAN

diabetest1<-read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")

A<-c(diabetest1$Age)

Mean<-mean(A)

MINIMUM

diabetest1<-read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")

A<-c(diabetest1$Age)

Minimum<-Min(diabetest1$Age)

MAXIMUM

diabetest1<-read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")

A<-c(diabetest1$Age)

Maximum<-Max(diabetest1$Age)

MINMAX

diabetest1<-read_excel("C:/Users/M.Geetha/Downloads/NARA.xlsx")

A<-c(diabetest1$Age)

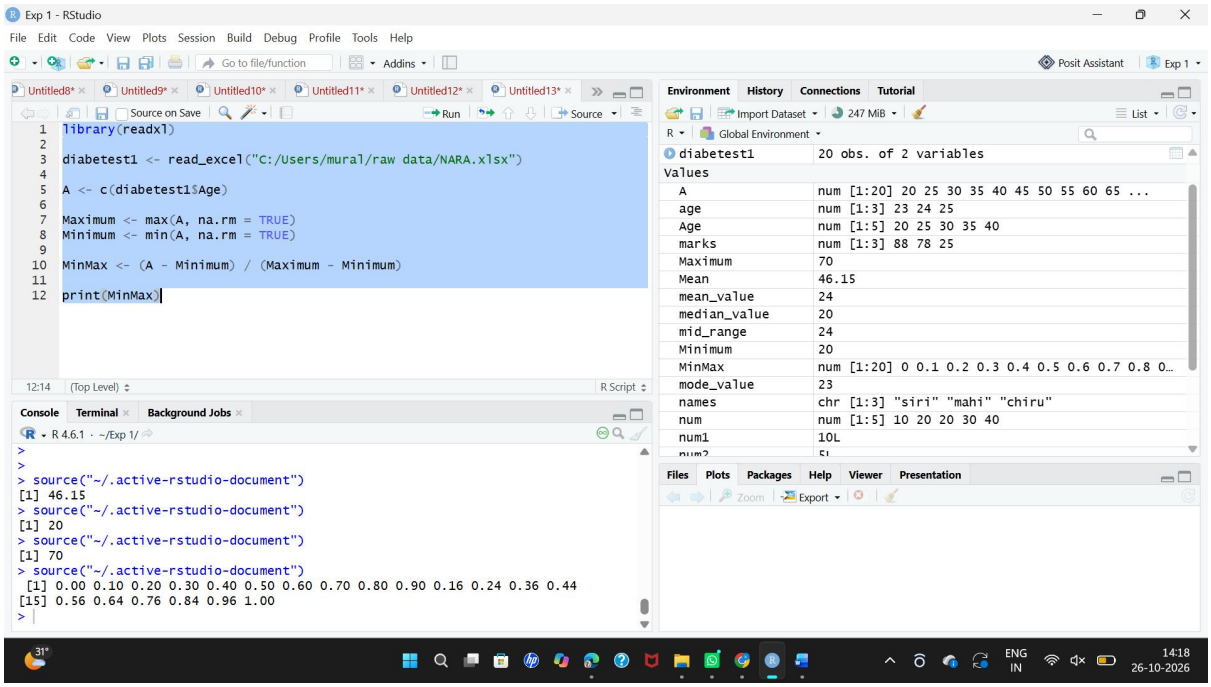
Maximum<-Max(diabetest1$Age)

Minimum<-Min(diabetest1$Age)

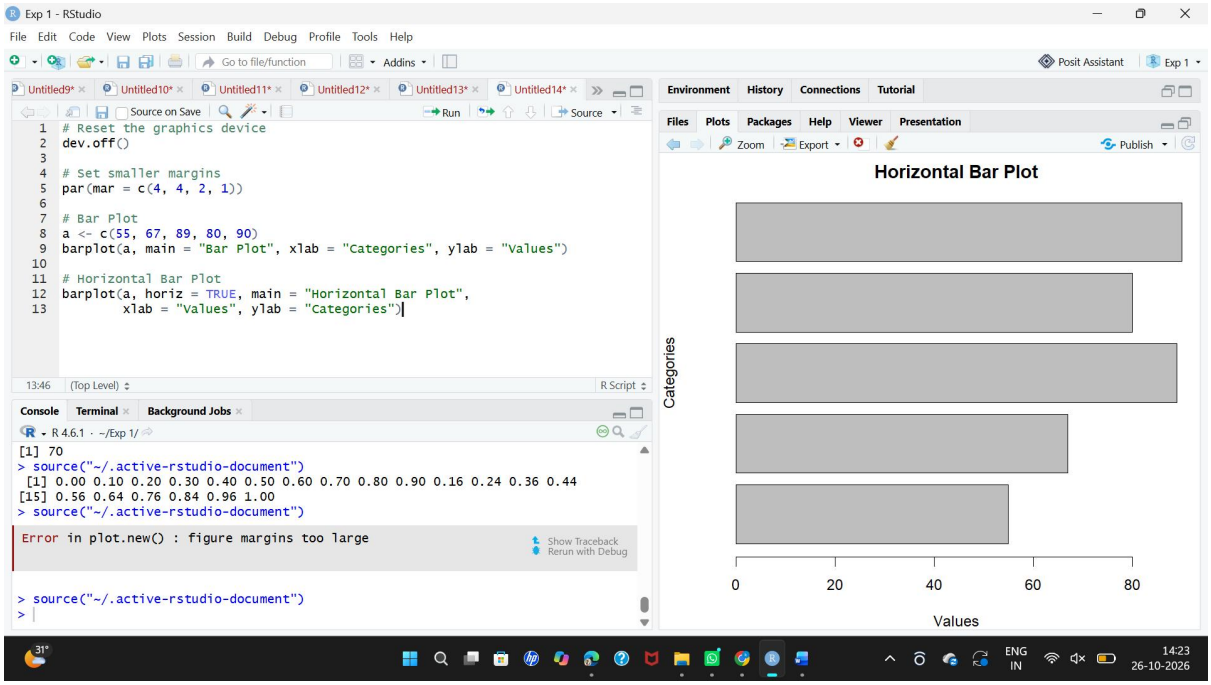
MinMax<-(A-Minimum)/(Maximum-Minimum)

MinMax

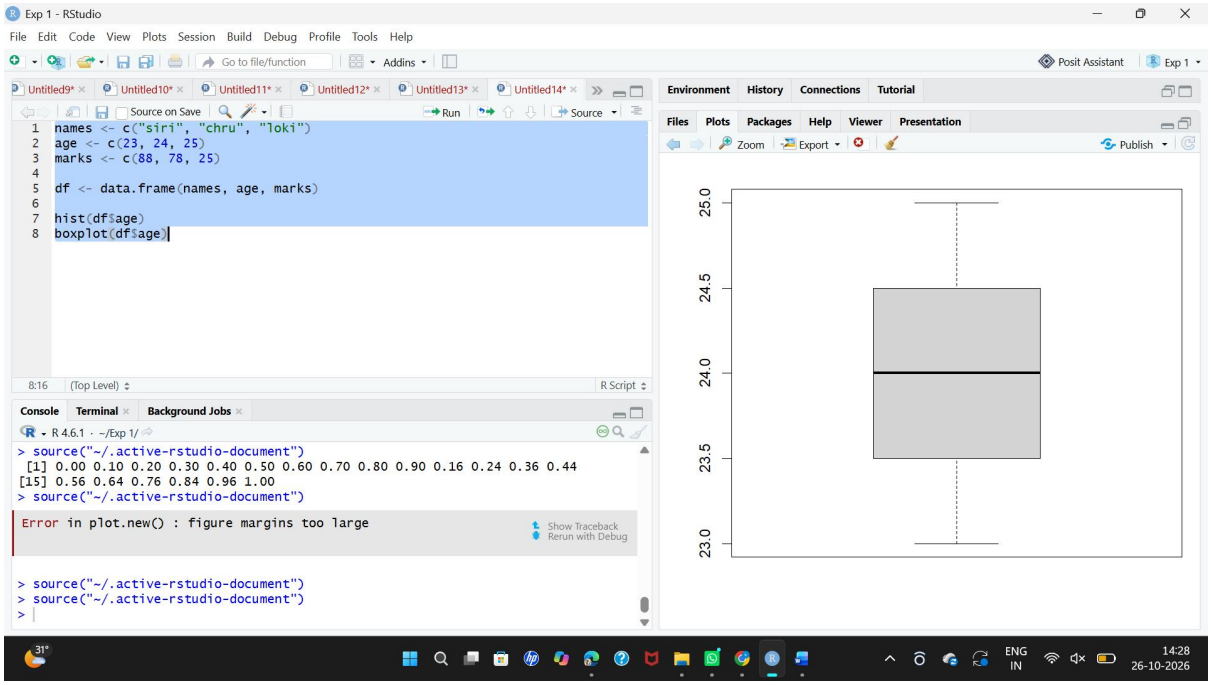
OUTPUT:


Exp 1 - RStudio
File Edit Code View Plots Session Build Debug Profile Tools Help
Untitled8* Untitled9* Untitled10* Untitled11* Untitled12* Untitled13*
Source on Save Run Source
1 library(readxl)
2
3 diabetest1 <- read_excel("C:/Users/mural/raw data/NARA.xlsx")
4
5 A <- c(diabetest1$Age)
6
7 Maximum <- max(A, na.rm = TRUE)
8 Minimum <- min(A, na.rm = TRUE)
9
10 MinMax <- (A - Minimum) / (Maximum - Minimum)
11
12 print(MinMax)
12:14 (Top Level) R Script
Console Terminal Background Jobs
R 4.6.1 · ~/Exp 1/
> source("~/.active-rstudio-document")
[1] 46.15
> source("~/.active-rstudio-document")
[1] 20
> source("~/.active-rstudio-document")
[1] 70
> source("~/.active-rstudio-document")
[1] 0.00 0.10 0.20 0.30 0.40 0.50 0.60 0.70 0.80 0.90 0.16 0.24 0.36 0.44
[15] 0.56 0.64 0.76 0.84 0.96 1.00
Environment History Connections Tutorial
Import Dataset 247 MiB List
R Global Environment
diabetest1 20 obs. of 2 variables
Values
A num [1:20] 20 25 30 35 40 45 50 55 60 65 ...
age num [1:3] 23 24 25
Age num [1:5] 20 25 30 35 40
marks num [1:3] 88 78 25
Maximum 70
Mean 46.15
mean_value 24
median_value 20
mid_range 24
Minimum 20
MinMax num [1:20] 0 0.1 0.2 0.3 0.4 0.5 0.6 0.7 0.8 0...
mode_value 23
names chr [1:3] "siri" "mahi" "chiru"
num num [1:5] 10 20 20 30 40
num1 10L
Files Plots Packages Help Viewer Presentation
Zoom Export
ENG IN
14:18
26-10-2026

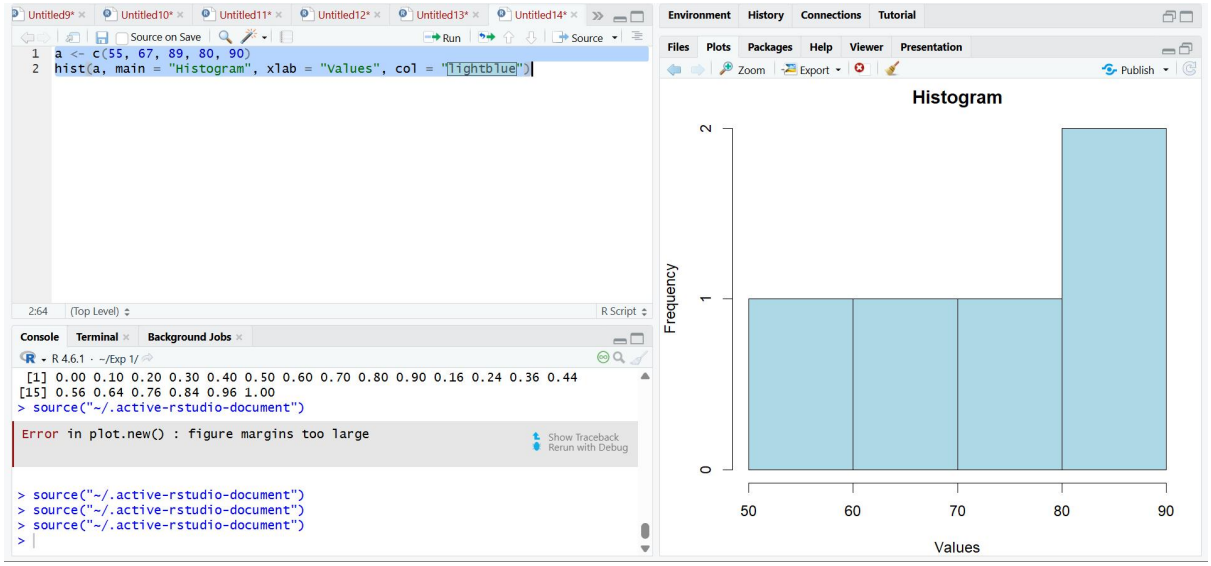
EXPERIMENT-14 BOX PLOT:



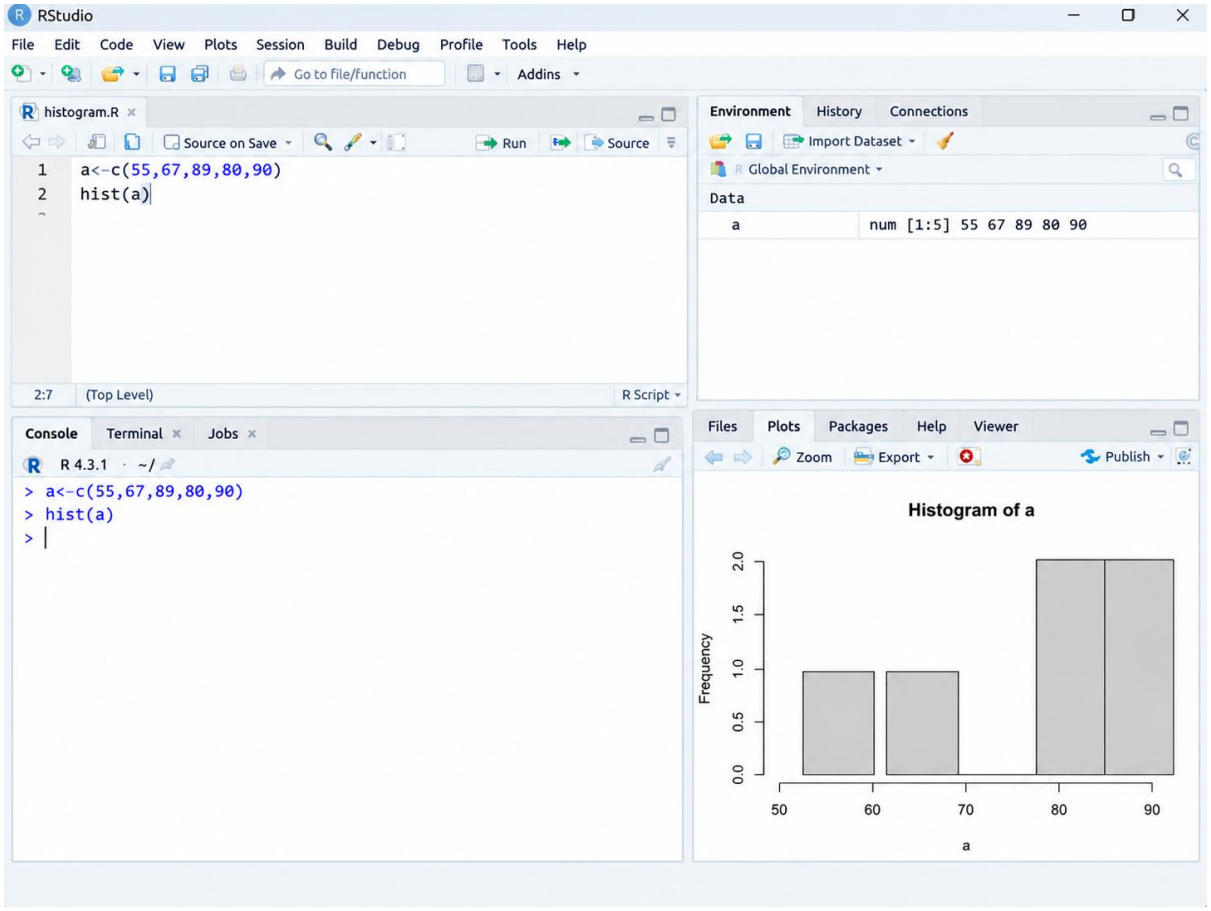
EXPERIMENT-15 BOX PLOT:



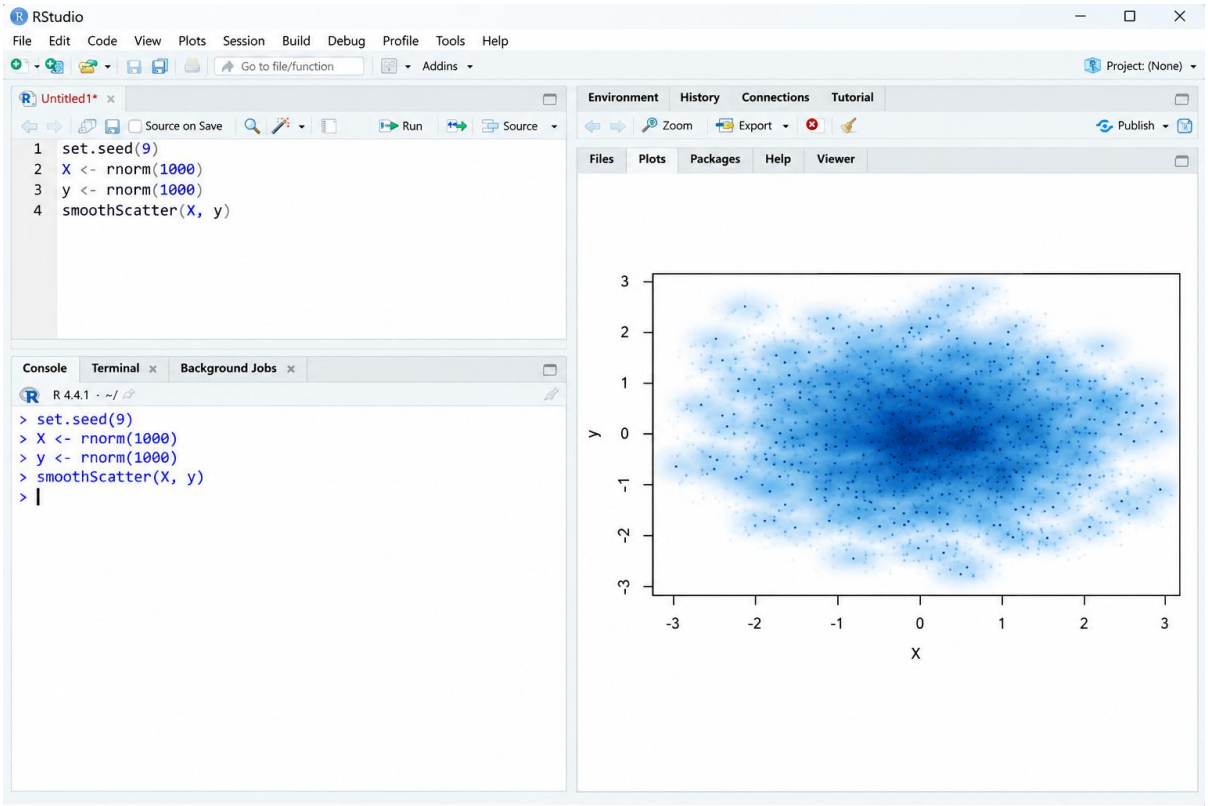
EXPERIMENT-16 HISTOGRAM:



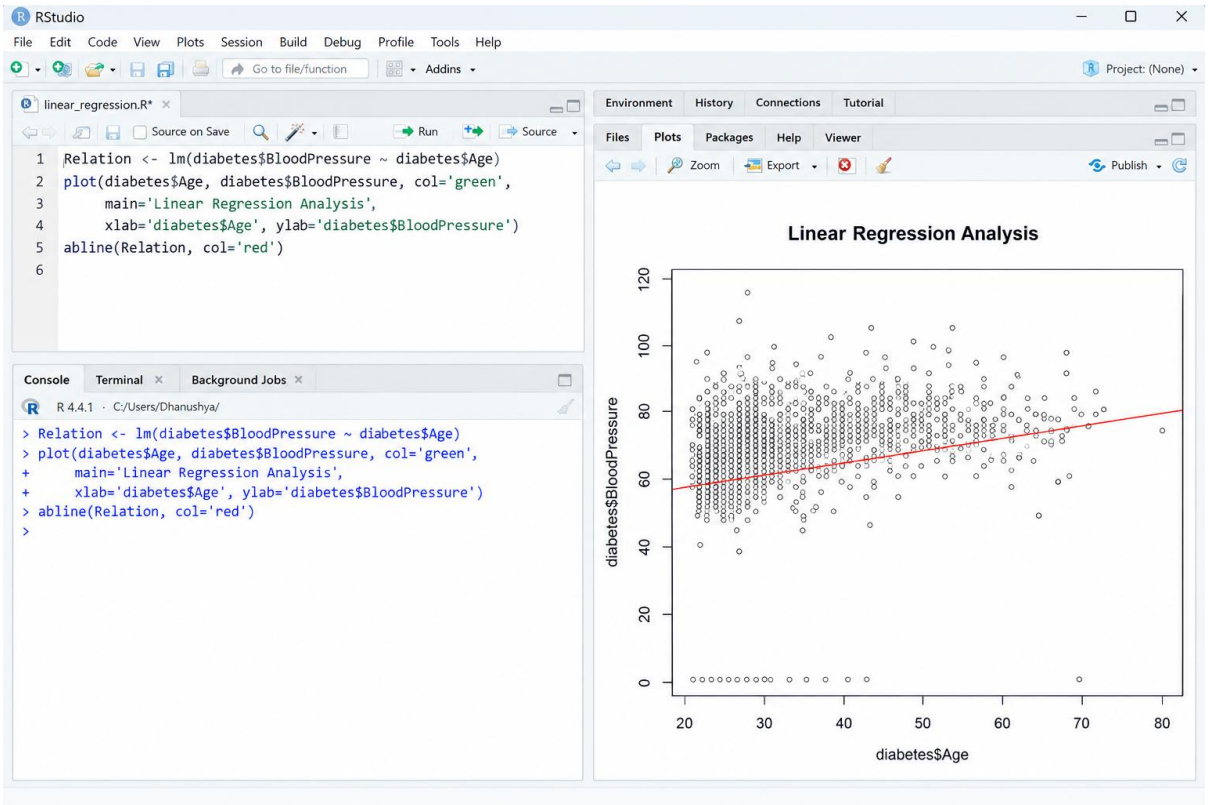
EXPERIMENT -17



EXPERIMENT -18



EXPERIMENT -19



EXPERIMENT 20

